

EXAM 7 – SPRING 2012

4. (3 points)

Given the following information:

Accident Year	<u>Incurred Loss Ratio</u>	
	As of 36 Months	As of 48 Months
2006	0.222	0.375
2007	0.451	0.675
2008	0.446	0.605
2009	0.228	

a. (1.5 points)

Estimate the loss ratio for accident year 2009 as of 48 months using the least squares method.

b. (1.5 points)

An alternate approach to estimating the accident year 2009 loss ratio as of 48 months is to use the arithmetic average of the link ratio method and the budgeted loss ratio method. Using the answer from part a. above, demonstrate whether this averaging approach is optimal.

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Question 4 Sample Answer

Solution 1

a)

$\underline{x}$	$\underline{y}$
0.222	0.375
0.451	0.675
0.446	0.605

$$L(x) = a + bx + \varepsilon$$

$$\hat{b} = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2} = \frac{(0.219) - (0.373)(0.552)}{(0.151) - (0.373)^2}$$

$$\begin{aligned} \bar{x} &= 0.373 & \overline{xy} &= 0.219 & \hat{b} &= 1.104 \\ \bar{y} &= 0.552 & \overline{x^2} &= 0.151 & \hat{a} &= \bar{y} - \hat{b}\bar{x} = 0.140 \end{aligned}$$

$$2009: L(0.228) = 0.140 + 1.104(0.228) = 0.392$$

b)

$$\text{Link ratio method: } L(x) = x \cdot \frac{\bar{y}}{\bar{x}} = \frac{x}{d} \quad \text{where } d = \frac{\bar{x}}{\bar{y}}$$

$$\text{Budget Loss: } L(x) = \bar{y} = EY$$

$$\text{cred-wtd: } L(x) = Z \frac{x}{d} + (1 - z) EY$$

$$\text{Arithmetic average} \Leftrightarrow Z = \frac{1}{2}$$

Least squares is equivalent to a credibility weighing between the link ratio method and the budget loss method with weight Z given to link ratio.

$$\text{where } Z = b \times d = (1.104) (0.373/0.552) = 0.746$$

Since arithmetic approach is equivalent to $Z = \frac{1}{2} \neq 0.746$ it is not optimal

Solution 2

a)

$$b = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2} = 1.175 \quad a = .1135 \quad y'(.228) = \boxed{.381}$$

$$\text{b) } L(x) = ZU_{LR} + (1-Z)U_0 \quad \text{LSE: } Z = \frac{b}{c} \quad \text{b from above, } c = \frac{\bar{y}}{\bar{x}} = 1.979$$

$$Z = .7944 > .5$$

A straight average approach would not give the chain ladder enough weight

Solution 3

a)

	As of 36	As of 48
AY	X	Y
06	0.222	0.375
07	0.451	0.675
08	0.446	0.605

$$b = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - (\bar{x})^2} = 1.175$$

$$a = \bar{y} - \bar{x}b = 0.1135$$

$$\% \text{ loss ratio} = 0.1135 + (1.175)(0.228) = \boxed{38.14\%}$$

b)

$$\text{LDF} = \frac{\bar{y}}{\bar{x}} = \frac{1.655/3}{1.119/3} = 1.479$$

$$\text{According to least square, other optimal } Z \text{ in } \frac{b/\bar{y}}{\bar{x}} = 1.175/1.479 = 0.794$$

By taking the arithmetic average of CL and budget loss ratio method, you are assuming that Z is 0.5. Thus, the alternate approach is NOT optimal.

Examiner Comment

The vast majority, over 95%, of candidates were able to apply the Least Squares credibility formulas to calculate a , b , and y_{48} . Errors, if any, were generally found in arithmetic errors

It was expected that the candidates would conclude that the arithmetic average of the two methods would not produce the optimal result. Full credit was given when the candidate recognized that the least squares method in (a) is the best linear approximation to the Bayesian approach with a credibility weighting of the two methods. The weight given to the chain ladder approach, would be $Z = \hat{b} / c$, where c is the weighted average LDF. The candidate should also recognize that the alternate approach is an approach where $Z = 0.50$.

Common errors which did not receive full credit for (b) involved the candidate calculating Z and proving that by using Z , you should arrive at the same answer as in part (a). These papers failed to actually answer the question asked about the optimality of the alternative approach.

Note that the actual calculation of the arithmetic average loss ratio need not be computed to receive credit for part (b).